

```

clc;

clear;

close all;

% -----
% Vehicle Parameters
% -----

m = 2000;      % mass (kg)
g = 9.81;      % gravity (m/s^2)
Cd = 0.36;     % drag coefficient
A = 2.9;       % frontal area (m^2)
rho = 1.2;     % air density (kg/m^3)
Cr = 0.020;    % rolling resistance coefficient
r = 0.36;      % wheel radius (m)
GR = 6;        % gear ratio
eta = 0.87;    % transmission efficiency
theta = 9*pi/180; % road gradient (9 degrees)

% -----
% Speed range
% -----

v = 0:1:100;   % speed in km/h
v_ms = v/3.6;  % convert to m/s

% -----
% Forces
% -----

F_rr = m*g*Cr*ones(size(v_ms)); % rolling resistance
F_ad = 0.5*rho*A*Cd.*v_ms.^2;    % aerodynamic drag
F_gr = m*g*sin(theta)*ones(size(v_ms)); % grade resistance

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% Total force

F_total = F_rr + F_ad + F_gr;

% -----

% Power and Torque

% -----

P = F_total .* v_ms;      % power (W)

T_wheel = F_total .* r;    % wheel torque

T_motor = T_wheel ./ (GR*eta); % motor torque

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% -----

% Graph 1 : Force vs Speed

% -----

figure;

plot(v, F_rr, '--');

hold on;

plot(v, F_ad, '--');

plot(v, F_gr, '--');

plot(v, F_total, 'LineWidth', 2);

grid on;

xlabel('Speed (km/h)');

ylabel('Force (N)');

title('Forces vs Speed');

legend('Rolling', 'Aerodynamic', 'Grade', 'Total');

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% -----

% Graph 2 : Power vs Speed

% -----

figure;

plot(v, P/1000, 'LineWidth', 2);

grid on;

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xlabel('Speed (km/h)');
ylabel('Power (kW)');
title('Power vs Speed');

% -----
% Graph 3 : Motor Torque vs Speed
% -----

figure;
plot(v, T_motor, 'LineWidth', 2);
grid on;
xlabel('Speed (km/h)');
ylabel('Motor Torque (Nm)');
title('Motor Torque vs Speed');

exportgraphics(f1,'Forces_vs_Speed.png','Resolution',300);

diary('EV_model_code.txt')
type EV_model.m
diary off

```